

Long-Tailed Recognition of Evidential Experts for Graph-level Classification

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Abstract

Graph-level classification involves analyzing the property of the whole graph, which is typically solved by using graph neural networks (GNNs). Existing efforts generally assume a balanced class distribution. However, real-world data often exhibit long-tailed distributions, i.e., tail classes have significantly fewer samples than head classes, and thus directly applying GNNs is eventually biased toward the head classes, resulting in limited generalization over the tail classes. Moreover, the predictions of existing algorithms are usually not trustworthy, and the trained classifiers remain ignorant to their predictive confidence. Towards this end, in this paper we develop a principled framework called GraphEVER for long-tailed graph-level classification. Technically, GraphEVER incorporates the beliefs of multiple experts and leverages the idea of subjective logic within the Dempster-Shafer Evidence Theory (DST). It can provide the evidence and uncertainty estimation for each expert, where the evidence is parameterized by a Dirichlet distribution to model class probability distribution, and the uncertainty is quantified via a well-defined theoretical framework. In this way, diverse experts can be integrated under DST to endow the classifier with both reliability and robustness. Moreover, we propose an evidence-based routing mechanism to dynamically assign experts, such that the tail classes can receive more attention, while the head classes can reduce redundant engaged experts, further cutting down the computational cost and improving the efficiency. Extensive experiments on seven datasets verify the superiority of our proposed framework.

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CCS Concepts

• Information systems → Data mining; • Mathematics of computing → Graph theory.

Keywords

Graph Classification, Imbalanced Learning, Multi-expert Learning

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1 Introduction

Graph-structured data, as a kind of ubiquitous data structure, have attracted significant attention in various promising applications including social media, biochemistry, academic citation and transportation. Mining on graphs allows us to discover latent patterns and has hence received widespread interests, covering a variety of tasks and domains. Among them, graph-level classification is a fundamental problem in graph data mining, which aims to analyze and predict the property of an entire graph, such as predicting the target properties of molecules [17, 30] and analyzing the biological functionality of compounds [23].

Recently, graph neural networks (GNNs) [15, 22, 25, 50] have propelled the development of graph-level classification, which have revealed impressive performance via incorporating the attributive and structural information. However, the extensive existing GNN methods generally assume that the number of the labels is in a balanced situation, i.e., the class distribution is balanced. In contrast, many datasets in real-world scenarios naturally exhibit highly-skewed class distributions [13], where a majority of classes (tail classes) contain a small number of labeled graphs, while few classes (head classes) contain abundant labeled graphs. Inevitably, directly applying GNNs on these class-imbalanced datasets is likely to occur

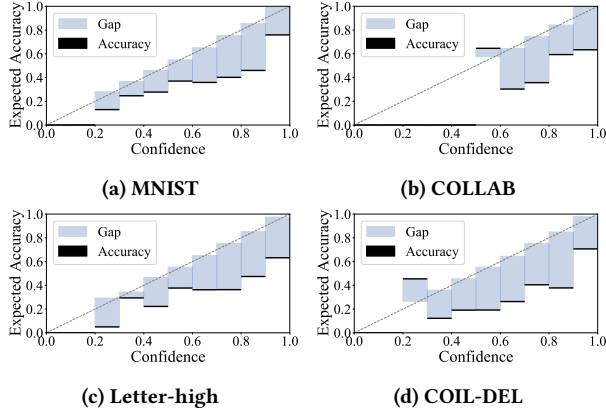


Figure 1: Reliability diagrams for GNNs trained on long-tailed data. The diagrams plot accuracy against confidence, where a perfect diagonal indicates ideal calibration, and any gap from this diagonal reflects model miscalibration.

the notorious *prediction bias* phenomenon [67], where extremely imbalanced classes bias the GNN classifier towards head classes, and tend to overfit the tail classes, resulting in poor generalization over the tail classes. It thus raises a meaningful question: *How do existing GNNs address severe class imbalances?*

Actually, there are a large number of proposed methods in vision domains to address class imbalances [64, 65], which can be categorized into three groups: *re-sampling*, *re-weighting*, and *ensembling learning*. Re-sampling strategies [4, 10, 66] aim at constructing synthetic training data to achieve a more balanced distribution. Re-weighting strategies [2, 6, 54, 62] adjust the portion of training loss of different classes by assigning weights to optimize the training objective. Different from the former, ensembling learning [24, 52, 55] combines multiple classifiers in a multi-expert framework to obtain reliable and robust predictions.

Although long-tailed learning has made relatively mature progress in the field of computer vision, for graph machine learning, we have identified three key challenges: **(i) Existing research on long-tailed graph-level classification has been scarcely explored.** Although some algorithms specifically designed for node-level classification on graphs have been proposed [29, 38, 39, 48, 49, 63], long-tailed learning for graph-level classification is another very practical and important issue. **(ii) Inability to estimate the uncertainty of the predictions.** GNNs contribute to the success of predictive accuracy. However, their predictions are not necessarily trustworthy. Studies have proved that traditional neural networks easily lead to over-confidence [9, 37]. Figure 1 presents the reliability diagrams for GNNs trained on long-tailed data. As illustrated in the diagrams, GNNs exhibit over-confidence in long-tailed scenarios, where predictions have high confidence but low accuracy. Hence, the model should know the predictive confidence in its judgment instead of making an incorrect one. **(iii) Most existing long-tailed algorithms often induce excessive computational resources.** Actually, re-sampling strategies typically address the class imbalance via augmenting plenty of synthetic training data, while ensembling learning strategies generally hold the assumption

that each classifier should undergo training on all available samples. For example, G^2 GNN [53] employs a topology-enhanced upsampling technique to generate balanced samples, leveraging a large number of pairwise samples for self-consistency regularization, which significantly increases the training burden. These limitations often result in redundant computational resources. Consequently, there is a compelling need to present an approach capable of estimating prediction uncertainty or confidence while concurrently alleviating excessive computational resources.

In this paper, we develop a novel framework named **Evidential ExpeRts** (GraphEVER) for long-tailed graph-level classification. The primary concept involves jointly predicting the classes and estimating the uncertainty of the predictions in a multi-expert framework. To achieve this goal effectively, our GraphEVER is built upon the decision support of the multiple experts to acquire the beliefs from a theory of evidence perspective [8, 14], where each expert uses the subjective logic to provide the evidence and uncertainty estimation based on the Dempster-Shafer Evidence Theory. Technically, the evidence captures the class probability distribution, parameterized using a Dirichlet distribution. And the uncertainty is quantified via a well-defined theoretical framework to express the opinion of “*I do not know*”. In this way, we can well measure the predictive confidence of the trained classifier. Moreover, to cut down the redundant computational resources, we develop an evidence-based routing mechanism to dynamically assign experts, such that tail classes are capable of receiving more attention to enhance the generalization, while head classes utilize fewer experts to maintain competitive performance, further reducing computational costs. Our extensive experiments demonstrate that it can largely improve the existing state-of-the-art methods on seven benchmark datasets.

2 Preliminary

Consider a set of N graphs $\mathcal{G} = \{G_i\}_{i=1}^N$, let $G_i = \{(\mathcal{V}_i, \mathcal{E}_i, y_i)\}$ denote a graph with its corresponding label $y_i \in \{1, 2, \dots, K\}$, where $\mathcal{V}_i$ is the set of nodes, $\mathcal{E}_i$ represents the set of edges, and K stands for the total number of classes. In addition, let n_i represent the count of graphs in the i -th class ($i = 1, 2, \dots, K$), with the assumption that $n_1 \geq n_2 \geq \dots \geq n_K$.

Definition 2.1 (Long-tailed Datasets). For a given dataset, when its classes are arranged in descending order based on their cardinalities, it qualifies as a long-tailed dataset if the distribution of the sorted classes adheres to Zipf’s law [36], i.e., $n_i = n_1 \times i^{-\mu}$, where μ serves as a hyper-parameter governing the degree of class imbalance. The **imbalance factor (IF)** quantifies this imbalance and is defined as n_1/n_K .

Definition 2.2 (Long-tailed Graph-level Classification). For a long-tailed graph dataset denoted as $\mathcal{G} = \{G_i, y_i\}_{i=1}^N$, the objective of long-tailed graph-level classification involves training an unbiased classifier $\mathcal{F}$, denoted by the mapping: $\mathcal{F} : \mathcal{F}(G_i) \rightarrow y_i$, such that the trained classifier can work well for graphs in both head and tail classes and generalize well on a balanced test dataset.

GNN Classifier. To derive an effective probability assignment for each input graph, we begin by leveraging the GNNs [18] to extract accurate information of feature attributes and structural topology. Specifically, let $f_\theta(\cdot)$ denote the GNN classifier with the network

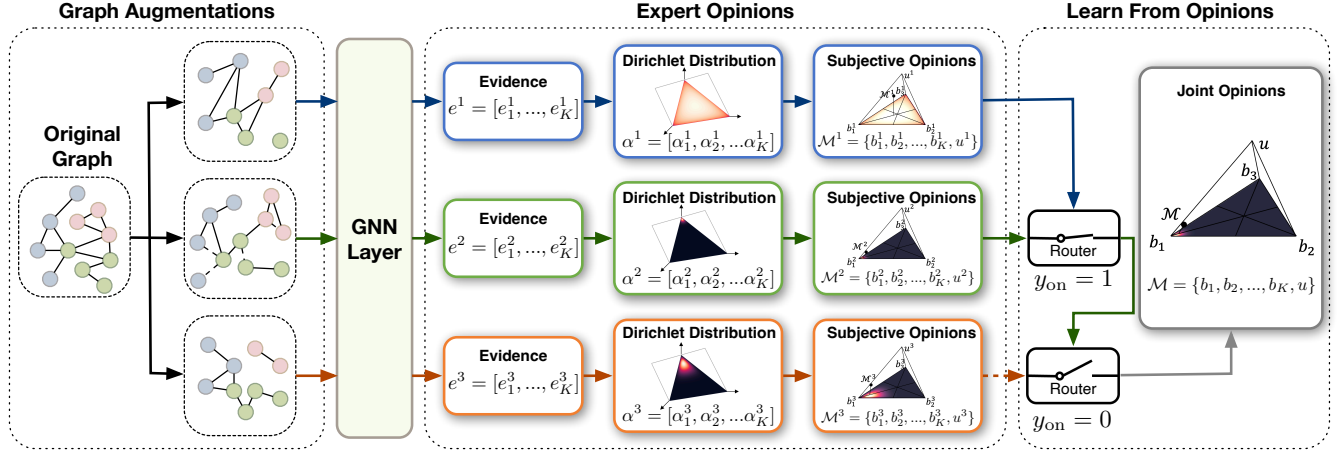


Figure 2: Overview of the proposed Evidential Experts (GraphEVER) framework for graph-level classification.

parameters θ , consisting of a L -layer GNN encoder and a multilayer perceptron (MLP) classifier, the propagation rule of in the l -th layer of GNN encoder can be expressed as:

$$\mathbf{h}_v^{(l)} = \mathcal{U}_\theta^{(l)} \left(\mathbf{h}_v^{(l-1)}, \mathcal{A}_\theta^{(l)} \left(\left\{ \mathbf{h}_u^{(l-1)} \right\}_{u \in \mathcal{N}(v)} \right) \right), \quad (1)$$

where $\mathbf{h}_v^{(l)}$ represents the node v 's representation at the l -th layer, and $\mathcal{N}(v)$ denotes the set of first-order neighbor nodes of node v . The functions $\mathcal{U}_\theta^{(l)}$ and $\mathcal{A}_\theta^{(l)}$ are associated with the updating and aggregation processes at the l -th layer. Hereafter, we can make further pooling operation [59] for the output node representations after L iterations, and summarize the graph representation $\mathbf{h}$ for graph G using the *READOUT* function:

$$\mathbf{h} = \text{READOUT}(\{\mathbf{h}_v^{(L)} : v \in \mathcal{V}\}), \quad (2)$$

Hence, the obtained graph representation proves valuable for subsequent graph-level classification tasks. Formally, we express the probability assignment for graph G as follows:

$$f_\theta(G) = \text{softmax}(\text{MLP}(\mathbf{h})). \quad (3)$$

However, the *softmax* function, while commonly used for estimating class probabilities in graph samples, has limitations. It tends to provide a point estimate, leading to potential over-confidence. Besides, the *softmax* function fails to provide the associated uncertainty. Therefore, our primary focus is on enhancing the accuracy of the GNN classifier and concurrently addressing the challenge of quantifying its uncertainty.

3 Methodology

In this paper, we introduce our framework GraphEVER for long-tailed graph-level classification. An illustration of the framework is presented in Figure 2.

3.1 Evidence-based Uncertainty Module

It has been shown that traditional neural networks easily suffer from over-confidence [9, 37]. This phenomenon is further exacerbated in the long-tailed classification, especially important in tail classes

with limited training samples, which shows the great uncertainty in the learning process.

In this section, we detail evidential deep learning to quantify the evidence confidence of each expert. This approach not only models the probability for each class but also captures the overarching uncertainty of predictions. Then we reduce the uncertainty of prediction by combining the subjective opinions of multiple experts through evidence theory. Finally, we elaborate on the training details of the learning objective.

3.1.1 Estimating Uncertainty for Each Expert. In probabilistic frameworks, the Dempster–Shafer Theory of Evidence (DST), or evidence theory, emerges as an extension of Bayesian principles with a focus on subjective probabilities [8]. DST employs belief functions to distribute belief masses across mutually exclusive sets of potential states, such as diverse class labels present in a sample. This framework enables the expression of uncertainty in predictions by acknowledging “*I do not know*” as a valid opinion [14, 42]. Subjective logic serves as the formalization tool for belief assignments and establishes a theoretical framework for acquiring the probabilities (belief masses) of different classes while addressing the overall uncertainty inherent in predictions.

Formally, for a classification task involving K categories, we establish K distinct and mutually exclusive singletons (such as class labels). Each singleton, denoted as $k = 1, 2, \dots, K$, is assigned a belief mass represented by b_k , and there is an additional mass termed the overall uncertainty mass denoted as u . These $K + 1$ mass values are strictly non-negative and collectively sum to one. This definition can be expressed as follows:

$$u + \sum_{k=1}^K b_k = 1, \quad (4)$$

where $b_k \geq 0$ for $k = 1, \dots, K$ and $u \geq 0$. For classification, the variables b_k and u are interpreted as the probability associated with the k -th class and the overall uncertainty. Additionally, let $e_k \geq 0$ denote the evidence corresponding to the belief mass b_k for a given singleton k . Then, the belief mass b_k and uncertainty u can be defined as:

$$b_k = \frac{e_k}{S}, \text{ and } u = \frac{K}{S}, \quad (5)$$

where $S = \sum_{i=1}^K (e_i + 1)$. For a more precise quantification of evidence, subjective logic associates the assignment of belief mass with a Dirichlet distribution, characterized by parameters $\alpha_k = e_k + 1$. Consequently, a subjective opinion can be constructed using the Dirichlet distribution parameters, expressed as $b_k = (\alpha_k - 1)/S$, where $S = \sum_{i=1}^K \alpha_i$ denotes the Dirichlet strength. The Dirichlet distribution can be formally defined as follows:

Definition 3.1 (Dirichlet Distribution). It arises as an extension of the Beta distribution to the multivariate domain and is parameterized by its K concentration parameters $\alpha = [\alpha_1, \dots, \alpha_K]$. The probability density function for the vector $\mathbf{p}$ is defined as follows:

$$D(\mathbf{p} | \alpha) = \begin{cases} \frac{1}{B(\alpha)} \prod_{i=1}^K p_i^{\alpha_i-1} & \text{for } \mathbf{p} \in S_K, \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

Here, S_K represents the K -dimensional unit simplex, defined as:

$$S_K = \left\{ \mathbf{p} \mid \sum_{i=1}^K p_i = 1 \text{ and } 0 \leq p_1, \dots, p_K \leq 1 \right\}, \quad (7)$$

and $B(\alpha)$ represents the K -dimensional multinomial beta function. As a practical tool, Dirichlet distribution can typically be regarded as the conjugate prior of the multinomial distribution [1].

Therefore, given a sample, the parameters of the Dirichlet distribution used for classification can be interpreted as evidence e_k for k -th class, and a Dirichlet distribution parametrized over evidence models second-order probabilities and uncertainty [14]. Specifically, we leverage the output of the GNN to construct the multinomial opinions. Formally, when presented with a graph sample G , the GNN classifier predicts the evidence vector for each class, denoted as $\mathbf{e} = [e_1, \dots, e_K]$. This prediction is achieved by substituting the *softmax* layer with a non-negative activation function, such as *ReLU*, as specified in Eq. (3). Consequently, the Dirichlet distribution parameters become $\alpha = f_\theta(G) + 1$, and the mean α/S can be computed as an estimation of the class probabilities.

In practice, to enhance the diversity of experts, we introduce graph augmentations [5] into our framework to encourage the discrepancy. In particular, in graph domains, GraphCL [60] introduces four types of graph transformations via augmenting topological and attributive information of the graphs, which are beneficial for enhancing the diversity of experts. Specifically, for each expert, the graph G undergoes stochastic graph augmentations denoted as $\mathcal{T}(\cdot|G)$. This process results in a semantically preserved augmented graph, denoted as $\hat{G}$, achieved by randomly selecting one of four augmentation strategies mentioned above. Then the parameters of the Dirichlet distribution can be recalculated as $\alpha = f_\theta(\hat{G}) + 1$. In this way, for each expert, we can measure the evidence-based uncertainty directly from the outputs of GNNs, and such evidence theoretically addresses the problem of over-confidence, which is especially effective for long-tailed recognition.

3.1.2 Combining Multiple Experts with DST. Having acquired evidence and uncertainty for each expert, each one may still have its inherent prediction preference. How to combine all available evidence to eliminate bias and yield more trustworthy predictions is a nontrivial problem. Actually, one reliable solution is the ensembling learning through multiple experts, and Dempster-Shafer Evidence Theory (DST) offers a viable approach to fuse evidence from diverse sources effectively [43], thereby mitigating uncertainty, defined as:

Definition 3.2. (Dempster's combination rule for two experts) The joint mass $\mathcal{M} = \{\{b_k\}_{k=1}^K, u\}$ is combined from the probability mass assignments of two experts $\mathcal{M}^1 = \{\{b_k^1\}_{k=1}^K, u^1\}$ and $\mathcal{M}^2 = \{\{b_k^2\}_{k=1}^K, u^2\}$ in the following manner:

$$\mathcal{M} = \mathcal{M}^1 \oplus \mathcal{M}^2. \quad (8)$$

More specifically, the calculation rule is formulated as:

$$b_k = \frac{1}{1-C} (b_k^1 b_k^2 + b_k^1 u^2 + b_k^2 u^1), u = \frac{1}{1-C} u^1 u^2. \quad (9)$$

Here, $C = \sum_{i \neq j} b_i^1 b_j^2$ represents a conflict factor quantifying the level of inconsistency between the belief masses provided by two experts. The normalization term is defined as the scale factor $\frac{1}{1-C}$.

In a multi-expert framework involving M experts, the combination of beliefs from various sources can be performed sequentially using Dempster's rule of combination, which can be formulated as:

$$\mathcal{M} = \mathcal{M}^1 \oplus \mathcal{M}^2 \oplus \dots \oplus \mathcal{M}^M. \quad (10)$$

In this way, we can obtain the joint mass $\mathcal{M} = \{\{b_k\}_{k=1}^K, u\}$ under DST. Accordingly, the corresponding collective evidence from multiple experts and the parameters of the Dirichlet distribution can be determined using Eq. (5):

$$S = \frac{K}{u}, e_k = b_k \times S, \text{ and } \alpha_k = e_k + 1. \quad (11)$$

Based on the above combination rule, the derived joint evidence $\mathbf{e}$ from multiple experts and the associated parameters of joint Dirichlet distribution α can be induced to yield the final probability for each class, along with the overall uncertainty in the prediction.

3.1.3 Learning to Form Opinions. Here we explore the training strategy that enables the model to learn evidence for each expert [24, 42]. Actually, by replacing the *softmax* layer with a non-negative activation function, such as *ReLU*, the model outputs can be treated as the evidence vector $\mathbf{e}$. This, in turn, allows for the determination of the Dirichlet parameters α . These results can be integrated via subjective logic to reflect the opinion of the expert, namely, the confidence and uncertainty of the prediction.

Technically, for a graph sample G_i along with its ground-truth class label y_i represented as a one-hot encoded vector. We first construct the Dirichlet distribution $D(\mathbf{p}_i | \alpha_i)$, which is a prior on the multinomial likelihood $Multi(\mathbf{y}_i | \mathbf{p}_i)$. Then we adopt the Type II Maximum Likelihood Estimation to formulate our loss function:

$$\begin{aligned} \mathcal{L}_i &= -\log \left(\int \prod_{j=1}^K p_{ij}^{y_{ij}} \frac{1}{B(\alpha_i)} \prod_{j=1}^K p_{ij}^{\alpha_{ij}-1} d\mathbf{p}_i \right) \\ &= \sum_{j=1}^K y_{ij} \left(\log(S_i) - \log(\alpha_{ij}) \right). \end{aligned} \quad (12)$$

The above loss indicates that the evidence of the ground-truth class is encouraged to be higher than other class labels. However, it cannot guarantee that incorrect labels would produce less evidence. To this end, we introduce a mechanism to drive the total evidence to approach zero for a graph sample that cannot be accurately classified. Formally, for each expert, we incorporate the KL-divergence $KL(\cdot)$ into the loss function:

$$\mathcal{L}_{exp} = \sum_{i=1}^N \mathcal{L}_i + \lambda_t \sum_{i=1}^N KL(D(\mathbf{p}_i | \tilde{\alpha}_i) \| D(\mathbf{p}_i | \mathbf{1})). \quad (13)$$

where $\lambda_t = \min(1, t/T) \in [0, 1]$, and t denotes the current training epoch, T refers to the annealing epoch. $D(\mathbf{p}_i|\mathbf{1})$ represents the uniform Dirichlet distribution, with $\mathbf{1}$ as the parameter vector consisting of K ones. Furthermore, we define $\tilde{\alpha}_i = \mathbf{y}_i + (1 - \mathbf{y}_i) \odot \alpha_i$, where $\odot$ signifies the Hadamard product. $\tilde{\alpha}_i$ is the Dirichlet distribution parameter which ensures that the evidence for the ground-truth class is not mistakenly considered as zero. Specifically, the KL-divergence term can be formulated as follows:

$$KL[D(\mathbf{p}_i|\tilde{\alpha}_i) \| D(\mathbf{p}_i|\mathbf{1})] = \log \left(\frac{\Gamma(\sum_{k=1}^K \tilde{\alpha}_{ik})}{\Gamma(K) \prod_{k=1}^K \Gamma(\tilde{\alpha}_{ik})} \right) + \sum_{k=1}^K (\tilde{\alpha}_{ik} - 1) \left[\psi(\tilde{\alpha}_{ik}) - \psi\left(\sum_{j=1}^K \tilde{\alpha}_{ij}\right) \right], \quad (14)$$

where $\Gamma(\cdot)$ and $\psi(\cdot)$ are the gamma function and digamma function.

Additionally, to make the evidence contained by multiple experts as diverse as possible, an intuitive approach is to compare different experts pairwise, hoping that their prediction results are as different as possible. However, this brings a risk: if two experts make completely opposite predictions, such conflict would contradict our goal of achieving more accurate predictions through a multi-expert framework, which is unreasonable. To address this, we utilize normalized Dirichlet parameters α_{ik}^m/S_i^m to form Dirichlet distribution $D(\mathbf{p}_i|\alpha_i^m)$, and then the diversity of experts can be achieved through KL-divergence as:

$$\mathcal{L}_{div} = -\frac{1}{M} \sum_{m=1}^M KL(P(\mathbf{p}_i | \alpha_i^m) \| P(\mathbf{p}_i | \bar{\alpha}_i)) \quad (15)$$

where $\bar{\alpha}_i = \sum_{m=1}^M \alpha_i^m / M$ denotes the averaged Dirichlet parameters, and M is the number of experts. Through this strategy, the fused evidence distribution obtained through parameter averaging is more reasonably analogous to a clustering center. We aim for each expert to be appropriately distant from the clustering center, maintaining their diversity without encouraging them to optimize in completely opposite directions. Instead, we hope that while maintaining their differences, they will be trained through the previously mentioned supervised loss Eq. (13) to maintain consistency in their prediction results as much as possible. This ensures that the predictions from the multi-expert fusion framework are superior to those of individual experts.

Finally, the overall loss function of the joint training objective in our multi-expert framework can be formulated as follows:

$$\mathcal{L} = \sum_{i=1}^M \mathcal{L}_{exp} + \mathcal{L}_{div}. \quad (16)$$

3.2 Dynamic Routing Module

From the above module we know that for all class labels, we use all the experts to estimate the evidence and uncertainty of the prediction. In other words, for each class, all experts are involved even for some easily predictable classes. We argue that for many easy samples (usually in head classes), we do not need all the experts engaged, which would induce redundant computational resources and affect the efficiency of the inference process [52].

To this end, we develop a routing mechanism to dynamically assign experts, such that the tail classes can receive more attention, while the head classes can reduce redundant engaged experts. Specifically, we train a router to allocate these experts in the order needed. For example, if the first m experts are in charge of a graph sample G , our model produces the joint evidence of each class from

the first to the m -th expert with Dempster's combination rule, and can be used to predict for classification. Then the router makes a binary decision y_{on} regarding the allocation of the $m+1$ -th expert. If the m -th expert makes an incorrect prediction, but one of the rest $M-m$ experts classifies correctly, the router should output $y_{on}=1$ and otherwise $y_{on}=0$. In this way, to well adapt to long-tailed recognition, head classes should assign fewer experts, while tail classes will allocate as many experts as possible.

We achieve this by proposing an evidence-based binary classifier to learn each router. Specifically, for the m -th expert, we concatenate its evidence vector $\mathbf{e}^m$ with the top- s ranked joint evidence vector $\mathbf{e}$ from the first to m -th expert derived from the joint mass $\mathcal{M} = \{\{b_k\}_{k=1}^K, u\}$ under DST according to Eq. (11), where $\mathcal{M} = \mathcal{M}^1 \oplus \mathcal{M}^2 \oplus \dots \oplus \mathcal{M}^m$. Then we project the double-long vector to a scalar by a fully connected layer $\mathbf{W}^{(m)}$. Importantly, this layer is independent between routers. Finally, we apply the Sigmoid function $S(x) = \frac{1}{1+e^{-x}}$ to obtain a real activation value in $[0, 1]$:

$$r(x) = S(\mathbf{W}^{(m)}[\mathbf{e}^m \| \mathbf{e}_{\text{top-}s\text{-components}}]), \quad (17)$$

and this routing mechanism dynamically controls the switch of the router, and can be optimized by a binary cross-entropy loss:

$$\mathcal{L}_{\text{router}} = -y_{on} \log(r(x)) - (1 - y_{on}) \log(1 - r(x)). \quad (18)$$

By this means, routers dynamically control the number of engaged experts. During inference, we adopt a simple thresholding mechanism (0.5): if $r(x) < 0.5$, the classifier uses the current joint evidence for the final decision, otherwise it proceeds to the next expert. Compared to directly averaging all expert outputs in the multi-expert framework, conflicts among experts may yield inaccurate predictions. Moreover, this situation may result in the allocation of more experts than necessary, which does not effectively reduce computational costs or improve inference efficiency. Our evidence-based dynamic routing mechanism with theoretically guaranteed DST fusion rules more reasonably reflects the prediction tendencies after multi-expert fusion and more efficiently estimate the number of experts to be allocated, minimizing computational resources to the greatest extent. We summarize the optimization algorithm for GraphEVER in Algorithm 1 in Appendix A.

4 Experiment

4.1 Experimental Setup

4.1.1 Datasets. We evaluate our GraphEVER and baselines on seven publicly graph datasets in various fields, including (a) vision dataset: MSRC_9 [35], Letter-high [40], Letter-low [40], COIL-DEL [40], (b) social networks: COLLAB [56], (c) synthetic dataset: Synthie [34], and (d) bioinformatics dataset: ENZYMES [41].

In the implementation, training datasets are transformed into long-tailed formats with varying IFs to strictly adhere to Zipf's law, whereas the validation and test datasets are kept balanced.

4.1.2 Compared Baselines. To highlight the efficacy of our proposed approach, we benchmark the GraphEVER against several leading baselines. We categorize these baseline methods into four distinct groups for a comprehensive comparison: (a) Techniques for data re-balancing: up-sampling [3]; (b) Cost-sensitive learning methods: CB loss [6] and LACE loss [33]; (c) Information augmentation methods: graph augmentation [61] and G²GNN [53]; (d)

Table 1: Long-tailed graph classification accuracy (%) on seven benchmark datasets, varying the degree of imbalance by two IFs. The top classification results are highlighted in boldface and the second-best results are indicated with an underline.

Model	COLLAB		Synthie		ENZYMES		MSRC_9		Letter-high		Letter-low		COIL-DEL	
	IF=10	IF=20	IF=15	IF=30	IF=15	IF=30	IF=5	IF=10	IF=25	IF=50	IF=25	IF=50	IF=10	IF=20
GraphSAGE	63.07	53.33	34.74	30.25	30.66	25.16	82.00	79.10	51.06	42.16	86.00	84.32	38.80	31.32
Up-sampling	72.33	70.25	35.25	33.50	32.33	28.50	83.20	78.50	53.62	44.20	88.48	86.72	39.20	26.96
CB loss	68.78	65.85	34.75	30.75	32.19	26.83	81.50	76.50	53.76	45.06	87.46	85.44	41.72	32.34
LACE loss	68.33	64.77	33.25	30.85	31.16	25.50	80.50	80.20	47.46	38.94	87.89	84.69	41.96	32.18
Augmentation	72.85	71.14	39.37	35.37	32.08	26.75	85.00	78.75	49.28	42.36	88.32	86.40	38.18	30.80
G ² GNN _n	73.94	71.89	38.08	27.94	35.00	29.17	88.57	85.04	<u>58.91</u>	<u>51.12</u>	89.49	<u>87.98</u>	38.32	27.98
G ² GNN _e	<u>74.50</u>	<u>72.76</u>	40.19	<u>37.53</u>	35.83	29.50	<u>90.28</u>	<u>86.25</u>	58.85	49.96	<u>89.84</u>	87.80	39.18	31.06
GraphCL	69.33	67.36	40.25	36.25	36.66	29.83	88.37	84.69	57.34	48.93	89.28	87.89	42.02	33.19
SupCon	69.25	67.14	<u>40.34</u>	37.25	<u>37.08</u>	<u>30.67</u>	89.44	85.01	57.29	48.93	89.12	87.36	<u>42.93</u>	<u>34.20</u>
GraphEVER	77.13	75.07	41.63	38.25	38.17	32.25	90.50	87.00	63.40	54.86	92.90	92.12	45.52	36.28
GraphEVER _{rout}	77.33	75.24	41.45	37.50	38.33	32.16	88.75	86.50	63.73	55.37	92.75	91.94	45.00	35.86
Improve $\uparrow$	+3.80%	+3.41%	+3.20%	+1.92%	+3.37%	+5.15%	+0.24%	+0.87%	+8.18%	+8.31%	+3.41%	+4.71%	+6.03%	+6.08%

Contrastive learning-based methods: graph contrastive learning (GraphCL) [60] and supervised contrastive learning (SupCon) [21].

For our GraphEVER, the detailed model and parameter settings is provided in the Appendix C.

4.2 Overall Evaluation

We assess the performance of our proposed GraphEVER along with competitive baselines for long-tailed graph classification. Table 1 presents the results across seven benchmark datasets, each characterized by varying levels of IFs. From these findings, we derive the following insights:

- An in-depth examination of accuracy across seven datasets indicates a significant decline in the performance of all methods as the imbalance in class distribution between head and tail grows. This observation points to a tendency for GNNs to experience a significant drop in effectiveness, leading to lower classification results in scenarios characterized by long-tailed distributions.
- Across the four groups of competitive baseline methods, it is generally observed that information augmentation strategies outperform re-balancing methods on the majority of datasets by leveraging extra information to enhance the representation of tail classes. However, it degrades severely in extreme imbalance situations, like the COIL-DEL dataset, which has an IF of 20 and over 60 classes with fewer than 3 training examples each. On the other hand, baselines based on contrastive learning tend to maintain consistent performance across different datasets.
- Overall, from the quantitative results, it can be observed that our framework GraphEVER and its variant GraphEVER_{rout} achieve the best performance compared to other competitive baselines on all seven datasets with varying levels of class imbalance. In particular, GraphEVER_{rout} outperforms the closest competitor on Letter-high with 8.18% with IF=25 and 8.31% with IF=50, which demonstrates the excellent capability of our framework for estimating the uncertainty and performing trustworthy long-tailed graph classification. However, it should be noted that the variant GraphEVER_{rout} may result in performance degradation under

Table 2: Ablation study of classification accuracy (%) for various variants on the Letter-high and ENZYMES datasets.

	$\mathcal{L}_{mle}$	$\mathcal{L}_{kl}$	$\mathcal{L}_{div}$	Aug	Letter-high	ENZYMES
M_1	✓				62.70	35.67
M_2	✓	✓			62.87	35.17
M_3	✓		✓		62.98	36.00
M_4	✓	✓	✓		63.16	36.21
M_5	✓	✓	✓	✓	63.40	38.17

particularly severe long-tailed scenarios as it dynamically adjusts the expert engagement for each sample.

4.3 Ablation and Sensitivity Studies

Here we further investigate the functionality of various components within GraphEVER via ablation studies. Additionally, we explore how GraphEVER's performance is influenced by changes in the hyper-parameter related to the number of experts, denoted as M .

4.3.1 Functionality of Objective Components. Given that the overall loss function of GraphEVER comprises two main components: $\mathcal{L}_{exp}$ ($\mathcal{L}_{mle} + \mathcal{L}_{kl}$), and $\mathcal{L}_{div}$, conducting ablation studies is crucial to assess the impact of each individual component and the graph augmentation strategies (Aug) on overall effectiveness. Specifically, we explore four different variants designed as follows:

- M_1 : Our base model, in which each expert is trained utilizing only the loss function $\mathcal{L}_{mle} = \sum_{i=1}^N \mathcal{L}_i$, as defined by Type II Maximum Likelihood Estimation in Eq. (12).
- M_2 : It is a variant in which each expert is optimized utilizing both $\mathcal{L}_{mle}$ and the KL-divergence term $\mathcal{L}_{kl}$, i.e., the total $\mathcal{L}_{exp}$.
- M_3 : It is another variant, in which $\mathcal{L}_{mle}$ is combined with $\mathcal{L}_{div}$ in order to diversify each expert.
- M_4 : The model that trains with the overall loss function of the joint training objective in our multi-expert framework.

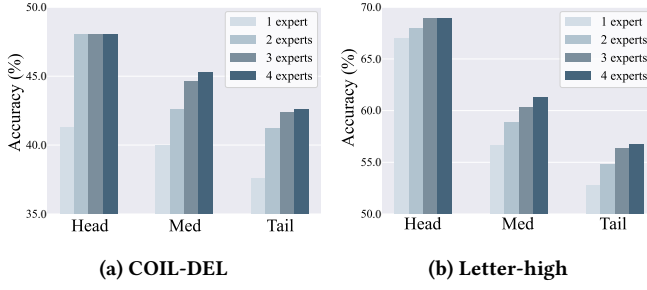


Figure 3: Performance comparison for the head, medium and tail classes w.r.t. various configurations of expert numbers.

- M_5 : Our complete model, which combines all three objective components along with the graph augmentations.

We choose two representative datasets, Letter-high and ENZYMEs to conduct the experiments. The results presented in Table 2 reveal that: (i) the incorporation of $\mathcal{L}_{kl}$ is beneficial in achieving more reliable uncertainty estimation, as it ensures that less evidence is generated for incorrect labels. (ii) The application of $\mathcal{L}_{div}$ in the optimization of the multi-expert framework leads to a remarkable improvement in overall performance, due to the diverse opinions formed by each individual expert. (iii) Moreover, the utilization of graph augmentations plays a crucial role in diversifying the views observed by each expert and encouraging discrepancy, leading to enhanced graph-level representations for classification.

4.3.2 Influence of Expert Number. We explore the effect of varying expert numbers on accuracy across head, medium, and tail class segments within the COIL-DEL and Letter-high datasets, represented in Figure 3. We first divide the classes into the head, medium, and tail based on the sample quantity, and the corresponding results are recorded as the count of experts M increases. An initial boost in accuracy for head classes is noted with an increase in M from 1 to 2, but additional increases in M yield negligible benefits. For the medium and tail classes, leveraging more experts results in an overall improvement in performance. However, as the M increases, the improvement becomes increasingly small. This observation supports the notion that easy samples (head classes) typically require fewer experts, suggesting an optimization potential by allocating experts more flexibly based on class complexity.

4.4 Efficiency on Expert Engagement

Here we examine the computational efficiency of the proposed dynamic routing module and provide visualizations of the engagement of experts for the head, medium, and tail classes, to demonstrate that our dynamic routing module effectively makes performance and efficiency trade-offs.

4.4.1 Computational Cost. We assess the computational expenses of our proposed GraphEVER in comparison to other baseline methods, focusing specifically on the large-scale COIL-DEL dataset. We record the total training time when the model converges and the inference time of 1000 validation epochs in Table 3. To investigate the computational cost of the different numbers of experts, we create five variants of GraphEVER by varying the number of experts from

Table 3: Comparison of computational expenses across various methods, focusing on total training duration (seconds) at model convergence, inference duration (seconds) over 1000 epochs, and GPU memory usage during training (MB).

Model	Train	Inference	GPU Memory	Accuracy
GraphSAGE	445.61	353.58	1176	38.80
Up-sampling	2183.42	362.23	1178	39.20
CB loss	463.35	356.28	1176	41.72
LACE loss	466.92	354.32	1176	41.96
Augmentation	2823.64	375.23	1178	38.18
G^2GNN_n	5212.39	373.57	1536	38.32
G^2GNN_e	5432.08	394.07	1536	39.18
GraphCL	1188.12	369.23	1176	42.02
SupCon	1142.24	368.93	1176	42.93
GraphEVER (1 expert)	913.09	345.34	1178	41.64
GraphEVER (2 experts)	1087.77	364.46	1180	42.96
GraphEVER (3 experts)	1215.41	388.02	1182	45.52
GraphEVER (4 experts)	1385.73	419.96	1186	45.93
GraphEVER (4 experts + router)	1399.32	359.43	1186	45.57

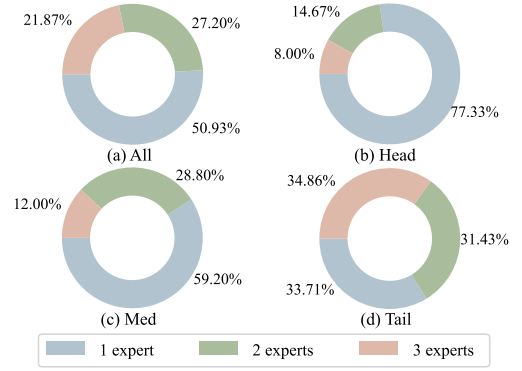


Figure 4: Visualization of expert utilization on Letter-high dataset through pie charts, which indicate the proportion of samples that involve a particular number of experts.

1 to 4. Additionally, we implement a dynamic router in the variant of GraphEVER that uses 4 experts. All baselines are trained and evaluated on a single NVIDIA A40 GPU. As is demonstrated in the table, our proposed GraphEVER has a lower computational cost in terms of total training time compared to prior data re-sampling (Up-sampling) and information augmentation (Augmentation, G^2GNN) methods, as these methods substantially increase the number of the input graphs through up-sampling. Moreover, G^2GNN introduces extra time complexity in the kernel similarity computation stage and message passing in the kNN graphs. As the number of experts increases from 1 to 4, both the training time and inference time increase accordingly, and we also observe a significant improvement in accuracy when increasing the number of experts. Moreover, the dynamic router effectively reduces the inference time by dynamically decreasing the engagement of experts for easy samples, while resulting in only a minor decline in overall performance. This finding validates the superiority of our dynamic routing mechanism. The GPU memory usage across all methods remains relatively consistent, indicating that incorporating multiple classifier heads as experts introduces affordable additional GPU memory overhead.

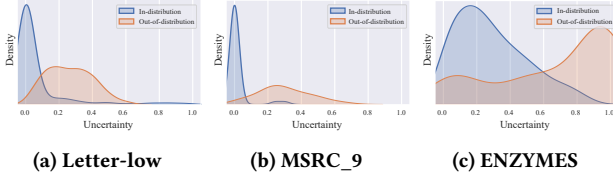


Figure 5: Uncertainty distribution measured by our model of in-/out-of-distribution samples.

4.4.2 Visualization of Expert Engagement. In Figure 4, we illustrate the distribution of expert usage among samples from head, medium, tail, and all class categories within the Letter-high dataset. We set the maximum expert count to 3, and leverage the routing mechanism to assign experts to different samples dynamically. As depicted in the figure, over half of the samples require only one expert to make confident decisions, suggesting the practicality of minimizing expert involvement. Furthermore, instances in tail classes tend to require more experts, while the majority of instances in head classes only require the first expert. The results validate the efficiency of our dynamic routing module in reducing unnecessary expert deployment for simpler cases, thereby improving computational efficiency. Moreover, the dynamic router allocates more experts to unconfident samples, resulting in improved performance.

4.5 Uncertainty Visualization

Here we assess the reliability of the estimated uncertainty by visualizing its distribution. Additionally, we present representative examples of the Dirichlet distribution to investigate the difference between the head and tail samples.

4.5.1 Uncertainty Estimation. We assess the reliability of the estimated uncertainty by illustrating how in-distribution and out-of-distribution samples are distinguished through their uncertainty scores. In-distribution samples are considered as those from the original dataset, whereas out-of-distribution samples are created by introducing Gaussian noise to the test samples, using a fixed standard deviation ($\sigma = 10$). This allows us to analyze the ability of the model to distinguish between in-distribution and out-of-distribution data. As depicted in Figure 5, the experimental results reveal the following observations: (i) Datasets with higher classification accuracy, such as Letter-low and MSRC_9, generally exhibit less uncertainty for in-distribution samples, while datasets with lower classification performance display greater uncertainty for these samples. (ii) Across all datasets, we observe significantly higher uncertainties for out-of-distribution samples, indicating that our model effectively characterizes uncertainty, thereby facilitating more reliable and robust discrimination between in-distribution and out-of-distribution data.

4.5.2 Case Study. To vividly demonstrate the Dirichlet distribution of head and tail samples under a triple classification task, we present representative examples in Figure 6 using the COLLAB dataset. The visualization shows the Dirichlet distribution of head and tail samples with varying numbers of experts. It is noted that the Dirichlet distribution associated with the head sample presents a concentrated distribution, with the density focused at the peak

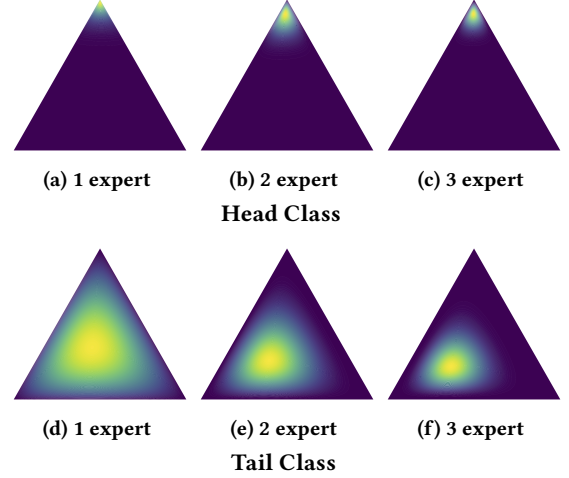


Figure 6: Case study of Dirichlet distribution.

of a standard 2-simplex. This indicates that sufficient evidence has been gathered for precise classification, resulting in low overall uncertainty. Moreover, it can be seen that a single expert is able to make confident decisions. In contrast, for tail classes, a single expert provides weak evidence, resulting in relatively high uncertainty and a flat distribution over the simplex. However, as more experts are engaged, the density of the distribution gradually shifts towards a specific direction, indicating that increasing expert engagement can improve the model’s ability to form confident decisions when single-expert evidence is insufficient for uncertain tail samples.

5 Conclusion

In this work, we explore an under-explored setting which so-called long-tailed graph classification. We propose a principled framework GraphEVER, which is built upon the decision support of the multiple experts to incorporate diverse beliefs from an evidence theory, and the subjective logic offers a methodology for integrating evidence and uncertainty assessments for each expert within the framework of Dempster-Shafer Evidence Theory. Further, to draw more attention to the tail classes and reduce redundant experts for the head classes, we develop a routing mechanism to dynamically assign experts. Extensive experiments and visualizations on seven datasets demonstrate the effectiveness of our GraphEVER.

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A Algorithm for our GraphEVER

Algorithm 1: Optimization Algorithm of GraphEVER

Input: Long-tailed graph dataset $\mathcal{G} = \{G_i, y_i\}_{i=1}^N$, number of experts M , number of classes K .

Initialize GNN classifier parameter θ and trainable weight matrices $\{\mathbf{W}^{(m)}\}_{m=1}^M$.

//Stage 1

while not converged do

for $m = 1 : M$ **do**

$\mathbf{e}^m \leftarrow$ the output of the m -th expert;

$\alpha^m \leftarrow \mathbf{e}^m + 1$;

$S^m \leftarrow \sum_{k=1}^K \alpha_k^m$;

$u^m \leftarrow K/S^m$;

 Compute the loss $\mathcal{L}^m$ of the m -th expert with Eq. (13);

end

 Compute the overall loss with Eq. (16) and minimize it to update parameters θ by gradient descent;

end

//Stage 2

while not converged do

 Compute the routing loss with Eq. (18) and minimize it to update parameters $\{\mathbf{W}^{(m)}\}_{m=1}^M$ by gradient descent;

end

Output: The joint evidence $\mathbf{e}$ with Eq. (11) for inference.

We show the pseudo-code of our proposed GraphEVER in the Algorithm 1.

B Related Work

B.1 Graph-level Classification

The development of graph representation learning algorithms [19, 26, 51] opens great opportunities for graph analysis. One core problem in the graph domain is graph-level classification, which involves predicting the category of the whole graph. Compared to node-level classification [27, 44, 45, 57] which targets the properties of individual nodes in a graph. Graph-level classification focuses on extracting more comprehensive representations of the graphs to make a better prediction. Early studies develop graph kernel methods [20, 46, 47] to solve the problem. Recently, graph neural networks [16, 31, 32, 58] emerge as the promising methods for this task, which have achieved unprecedented success in identifying categorical labels of graphs. To step further, we study a challenging scenario where the data are long-tailed, and we are the first to incorporate uncertainty estimation into this task.

B.2 Long-Tailed Recognition

Class-imbalanced learning (also known as long-tailed recognition) is studied actively in the vision domain. Existing approaches can be broadly categorized into re-sampling [4, 10, 66], re-weighting [2, 6, 54, 62], and ensembling learning [24, 52, 55]. Re-sampling methods seek to achieve a balanced distribution where tail classes are over-sampled, whereas head classes are undersampled. Re-weighting

methods allocate different importance weights on different classes to adjust the portion of training loss. The first two strategies typically put more emphasis on the tail classes and improve overall performance at the expense of the head classes. Recently, ensembling learning methods have empirically shown stronger generalization, which combine multiple classifiers in a multi-expert framework. However, this strategy is typically prone to induce redundant computational resources.

Recently, diverse research endeavors has been dedicated to mitigating long-tail recognition challenges in graph-based tasks [28, 29, 38, 39, 48, 49, 63], which have achieved promising achievements. The main idea of these approaches is to either design degree-specific transformations on nodes or utilize structural features. Distinct from these methods that study long-tailed node-level classification, our proposed framework GraphEVER goes further and concentrates on under-explored and promising long-tailed graph-level classification, and innovatively explores this task from the perspective of predictive uncertainty.

B.3 Evidence Theory

The mathematical foundation known as the Dempster-Shafer Evidence Theory (DST) is introduced by Dempster to address uncertainty reasoning [7]. Acting as an extension of the Bayesian theory to subjective probabilities [8], DST provides the opinion of both imprecision and uncertainty. It has been proposed as a more flexible and general approach than the Bayesian one. It stands out for combining the measures of evidence of multiple sources [11, 12, 43]. Benefits from its theoretical guarantee and flexible capability, our proposed work introduces this technique to accurately provide uncertainty estimation and predictive confidence directly.

C Model Settings

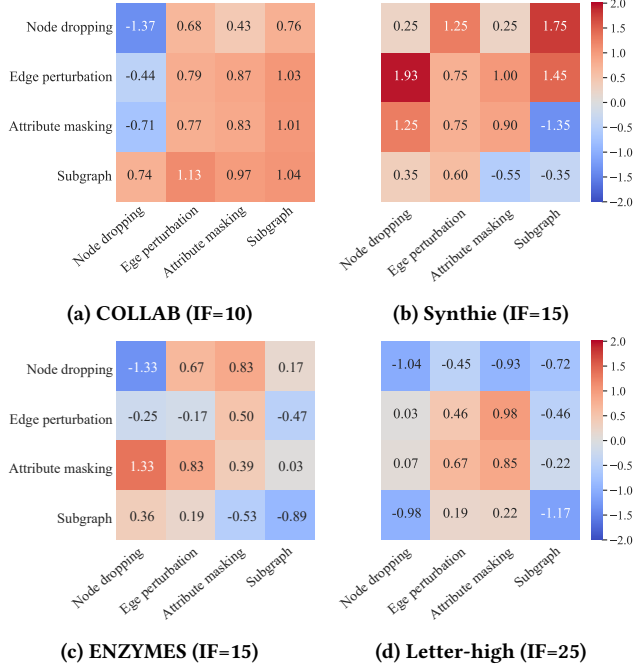
In our assessment of the GraphEVER and the various baselines, we employ GraphSAGE as the foundational GNN encoder and adjust the embedding size to 64. Adam optimizer is adopted for optimizing all models, setting the batch size to 32 and the learning rate to 0.0001. For our GraphEVER, the expert count M is set to 3 for performance and efficiency trade-off. Moreover, we adjust the hyper-parameters of annealing epoch T and top choices number s for each dataset. The re-weighting hyper-parameter β_{CB} for the CB loss is set to 0.99 and the scaling temperature t_{LACE} for the LACE loss is set to 1.0. Additionally, we fix the number of training epochs at 1000, and we implement the early stopping technique in the encoder training process with a patience of 500 epochs. In our experimental study, we employ accuracy as the standard metric to assess performance.

D Generalization beyond graph domain

To verify whether our proposed framework maintains its effectiveness beyond the graph domain, we evaluate our method on the well-known MNIST dataset from the computer vision field, with results presented in Table 4. As can be seen, our proposed method, GraphEVER and its variants GraphEVER_{route}, consistently outperform the baseline methods by a significant margin. This not only demonstrates the effectiveness of our approach in long-tail graph classification tasks but also shows that even when transferred to other domains, its performance remains competitive, highlighting the generalizability of our technical framework.

Table 4: Accuracy (%) on the MNIST datasets, varying the degree of imbalance by two IFs. The top classification results are highlighted in boldface and the second-best results are indicated with an underline.

	GraphSAGE	Up-sampling	CB loss	LACE loss	Augmentation	G^2GNN_n	G^2GNN_e	GraphCL	SupCon	GraphEVER	GraphEVER _{rout}
IF=50	68.67	64.69	68.85	69.72	69.37	69.76	72.18	70.91	<u>73.69</u>	75.38	75.62
IF=100	63.46	59.78	63.40	64.59	65.12	64.88	68.17	66.73	<u>70.31</u>	72.71	72.93

**Figure 7: Long-tailed graph classification accuracy gain (%) when using different graph augmentations. The accuracies of baselines training with no augmentations are 76.00%, 39.70%, 36.84%, 62.42% for the four datasets respectively.**

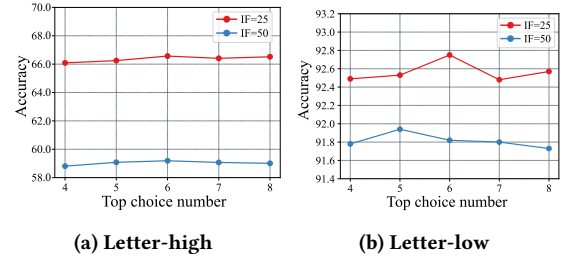
E Analysis of Graph Augmentations

In this section, we analyze the effectiveness of different types of graph augmentations on various datasets.

- **Node dropping.** It involves randomly removing a certain proportion of nodes and all connected edges, preserving semantics. The dropping probability follows an i.i.d. uniform distribution.
- **Edge perturbation.** It involves randomly adding or deleting a certain ratio of edges to alter the connectivity pattern.
- **Attribute masking.** It involves randomly selecting certain nodes and masking some of their attributes.
- **Subgraph.** It involves utilizing the random walk algorithm to extract a representative subgraph from the original graph, assuming that the semantic meaning of the graph is preserved.

We first set the number of experts to 3, and then adopt different types of augmentations for the input of the second and the third experts. The augmentation ratios for the second and the third expert are set to 0.05 and 0.1, respectively. In Figure 7, we demonstrate the accuracy of long-tailed learning gains when using different pairs of graph augmentations, where the warmer colors indicate better performance gains. As can be seen from the figure, using appropriate graph augmentations can benefit the classification performance significantly, indicating that graph augmentations improve model generalization by further diversifying and enriching the training data. It should be noted that the appropriate augmentation pairs vary among different datasets.

F Sensitivity Analysis

**Figure 8: Variations of the long-tailed graph classification accuracy of the GraphEVER under different settings of s .**

The hyper-parameter of the top choice number s need to be determined for the router training in GraphEVER. This section studies the parameter sensitivity of the method GraphEVER to parameter variations of s in terms of the long-tailed graph classification accuracy. The experiments are conducted on the Letter-high and Letter-low datasets under different imbalance settings. Figure 8 shows that the long-tailed graph classification accuracy of GraphEVER is not sensitive to parameter variations on the Letter-high dataset. For the Letter-low dataset, the accuracy under different values of s differs by no more than 0.4%. This indicates that the competitive performance of GraphEVER can be easily obtained by fine-tuning the hyper-parameters over a limited search space.